

Lattice EPID with Efficient Revocation

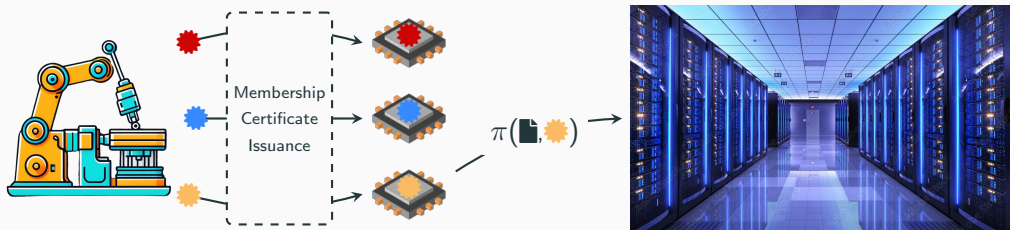
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¹ Orange, Applied Crypto Group

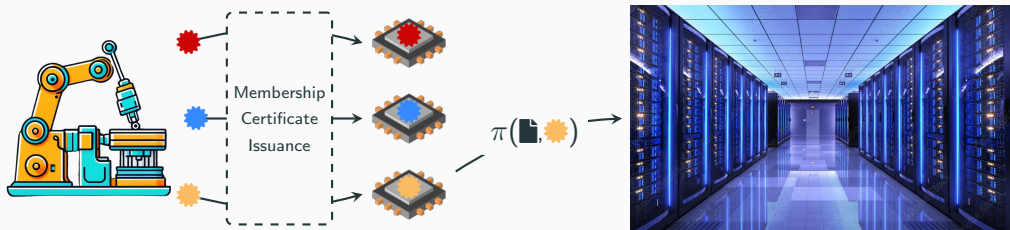


EPID: Group Attestation with Built-in Revocation Mechanism



Emphasis on **anonymity**, **unlinkability**, **revocability**

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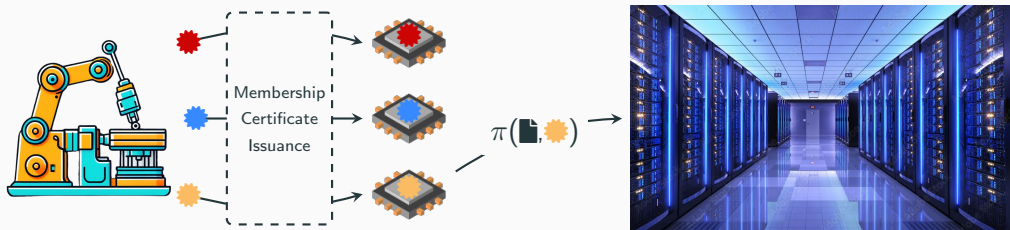
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Group Signatures

Trusted entity able to lift
anonymity of every signature

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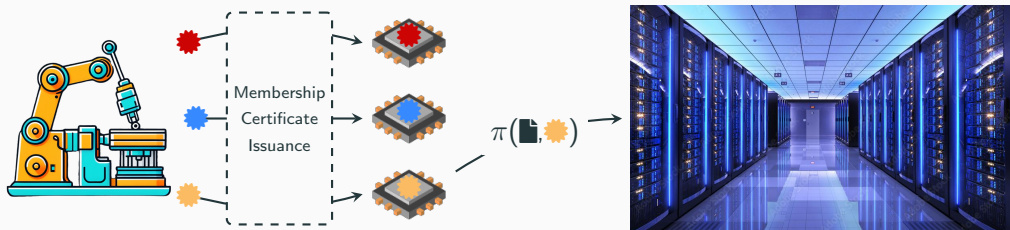


TPM

DAA

User-controlled linkability steered by a basename

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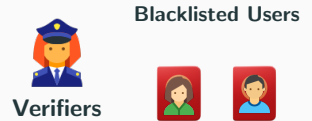
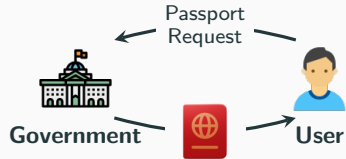


EPID

Platform revoked by adding one of its signature on a revocation list (ISO standard, deployed)

Intel SGX

How Is Revocation Done?



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Mainstream approach
Accumulators with (zero-knowledge) proof of accumulation / non-accumulation

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Centralized revocation: need a trusted entity to maintain the accumulator

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Mainstream approach

Accumulators with (zero-knowledge) proof of accumulation / non-accumulation

- ⊗ **Centralized revocation:** need a trusted entity to maintain the accumulator
- 🎭 **Who in the accumulator to revoke from an anonymous signature?**
 - 📁 Opening by a trusted entity (weaker anonymity guarantees)
 - 👤 Covering specific revocation scenarios, e.g., loss of credential signaled by users themselves

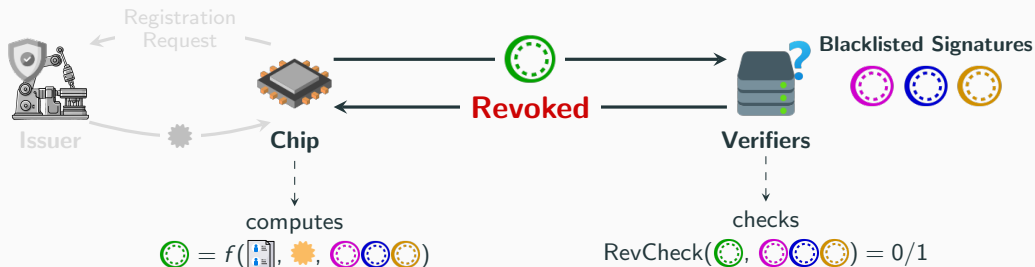


EPID anonymous revocation instead handled with **Signature Revocation Lists (SRL)**

- New token for each authentication, with proof user never produced blacklisted signatures

¹Sanders, Traoré. EPID with Malicious Revocation. CT-RSA 2021

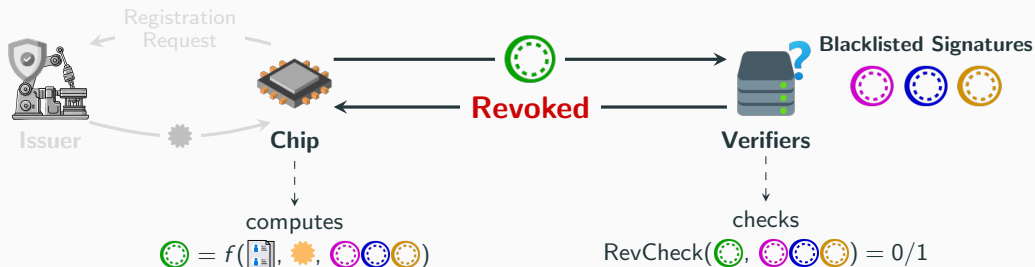
EPID Revocation In A Nutshell



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- New token for each authentication, with proof user never produced blacklisted signatures
- Verifier checks if user is revoked from anonymous token  and blacklisted ones 

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EPID anonymous revocation instead handled with **Signature Revocation Lists (SRL)**

- New token for each authentication, with proof user never produced blacklisted signatures
- Verifier checks if user is revoked from anonymous token (green circle) and blacklisted ones (purple, blue, orange circles)

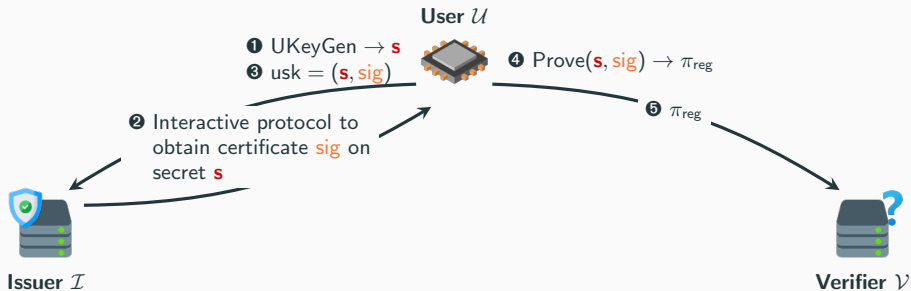


Better Trust Model

- No need for opening anonymous signatures
- Decentralized revocation: each verifier maintains its own SRL (possibly malicious [ST21]¹)
- Easy handling of problematic revocation scenarios like misbehaving users

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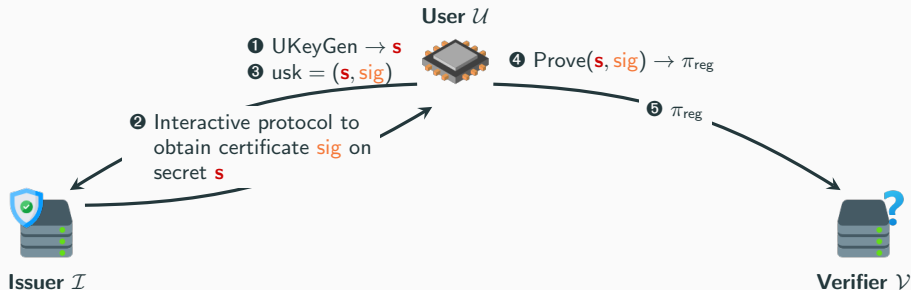
Starting from Group Signatures: Proving Membership



²Jeudy, Sanders. Worst-Case Lattice Sampler with Truncated Gadgets and Applications. Asiacrypt 2025

³Lyubashevsky, Nguyen, Plançon. Lattice-Based Zero-Knowledge Proofs and Applications: Shorter, Simpler, and More General. Crypto 2022

Starting from Group Signatures: Proving Membership



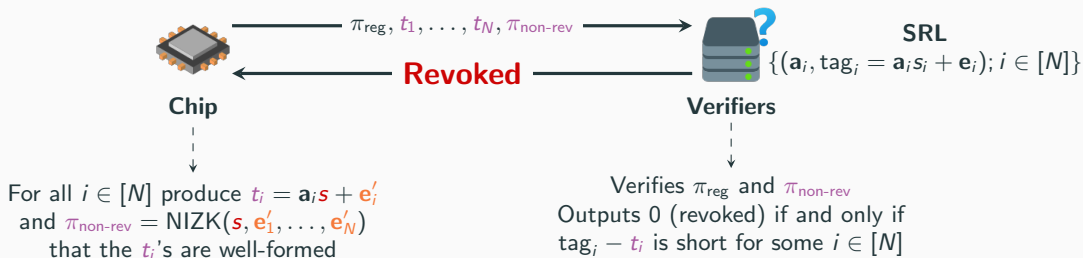
Well-Known Framework. Can be done efficiently with lattices [JS25]²:

- 2** $\mathcal{U}$ commits to $\mathbf{s}$ ($\mathbf{c} = \mathbf{A}\mathbf{r} + \mathbf{D}\mathbf{s}$)
 $\mathcal{I}$ chooses tag $\mathbf{t}$ and samples Gaussian preimage $\mathbf{v}$ of $\mathbf{u} + \mathbf{c}$ for $\mathbf{A}_{\mathbf{t}} = [\mathbf{A} | \mathbf{t}\mathbf{G}_H - \mathbf{A}\mathbf{R} | \mathbf{A}_3]$ (i.e., $\mathbf{A}_{\mathbf{t}}\mathbf{v} = \mathbf{u} + \mathbf{c}$)
 $\mathcal{U}$ computes $\text{sig} = (\mathbf{t}, \mathbf{w} = \mathbf{v} - [\mathbf{r} | \mathbf{0}])$, i.e., $\mathbf{A}_{\mathbf{t}}\mathbf{w} = \mathbf{u} + \mathbf{D}\mathbf{s}$
- 4** NIZK of $\mathbf{t}$ and short $\mathbf{s}, \mathbf{w}$ s.t. $\mathbf{A}_{\mathbf{t}}\mathbf{w} - \mathbf{D}\mathbf{s} = \mathbf{u}$ using [LNP22]³

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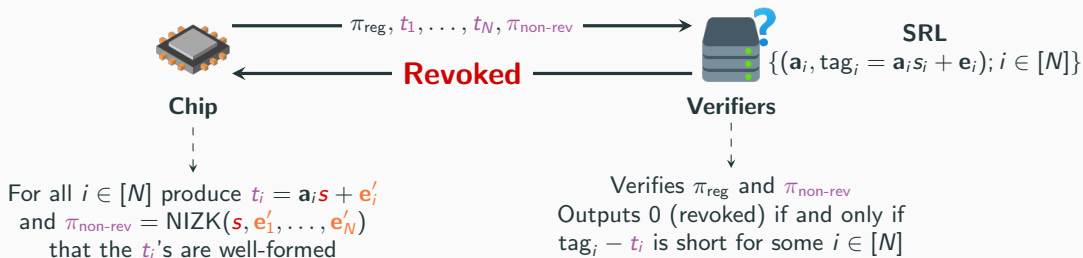
Adding Signature Revocation Lists



✓ **Correct**

$\text{tag}_i - t_i = a_i(s_i - s) + \bar{e}_i$ which is short only if $s = s_i$ (with overwhelming probability)

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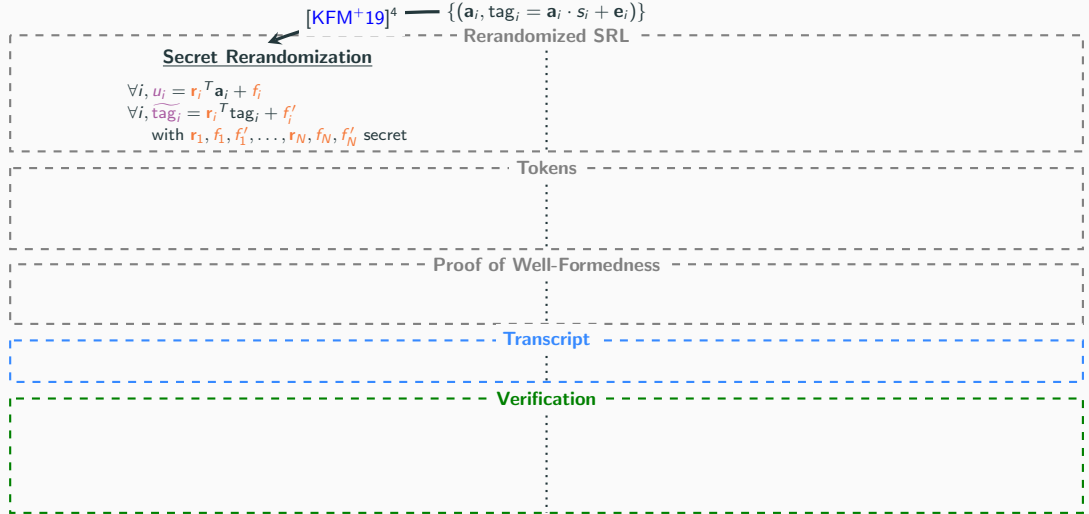
✓ **Correct**

$\text{tag}_i - t_i = a_i(s_i - s) + \bar{e}_i$ which is short only if $s = s_i$ (with overwhelming probability)

✗ **Insecure**

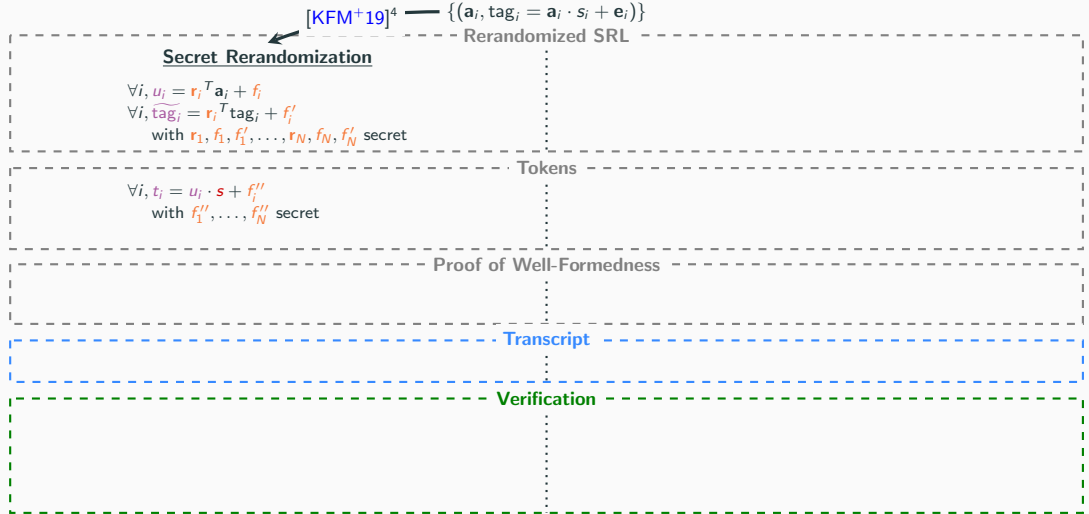
Authenticating twice for the same SRL or for intersecting SRL's would lead to $t_i = a_i s + e'_i$ and $t'_i = a_i s + e''_i$, which are close, allowing tracing the user.

Solution: SRL Randomization



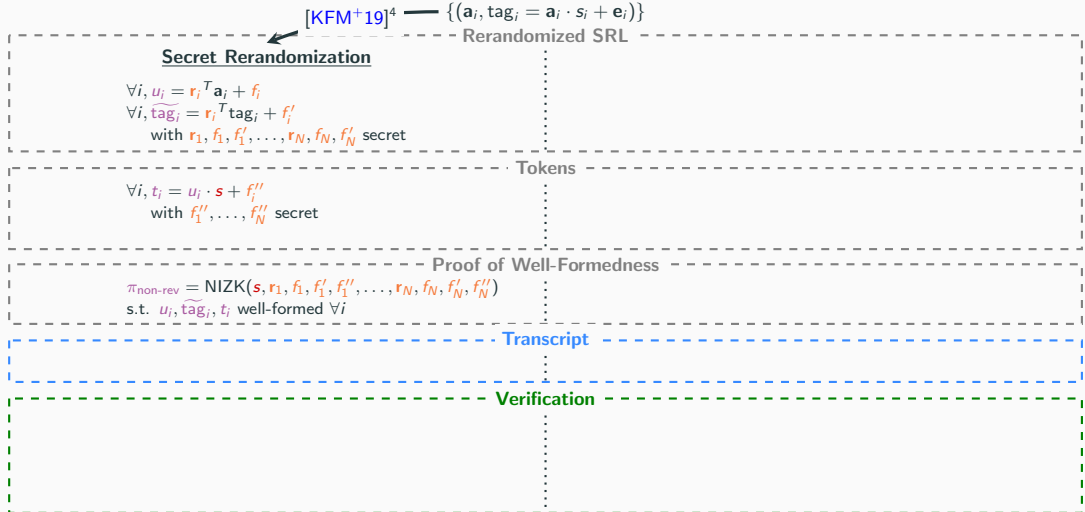
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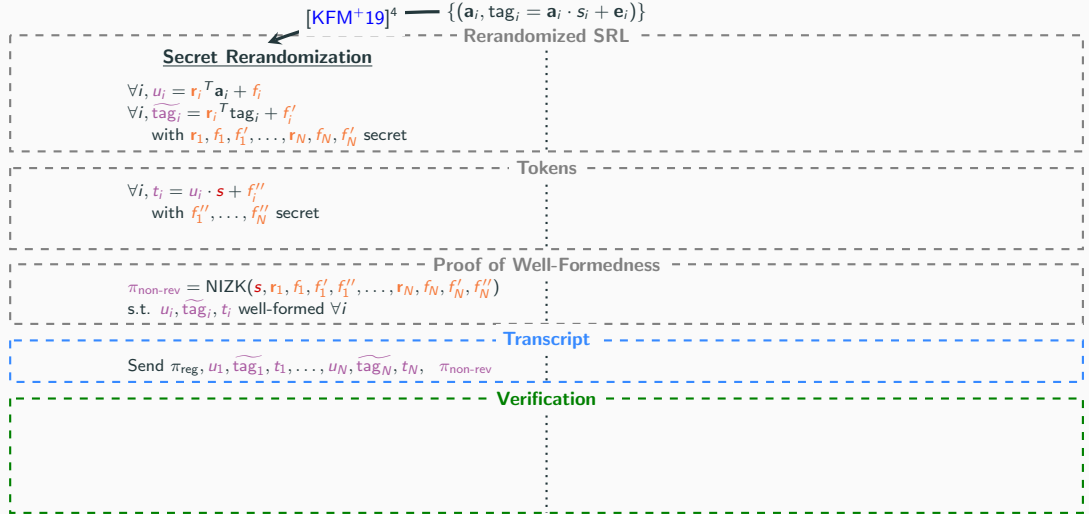
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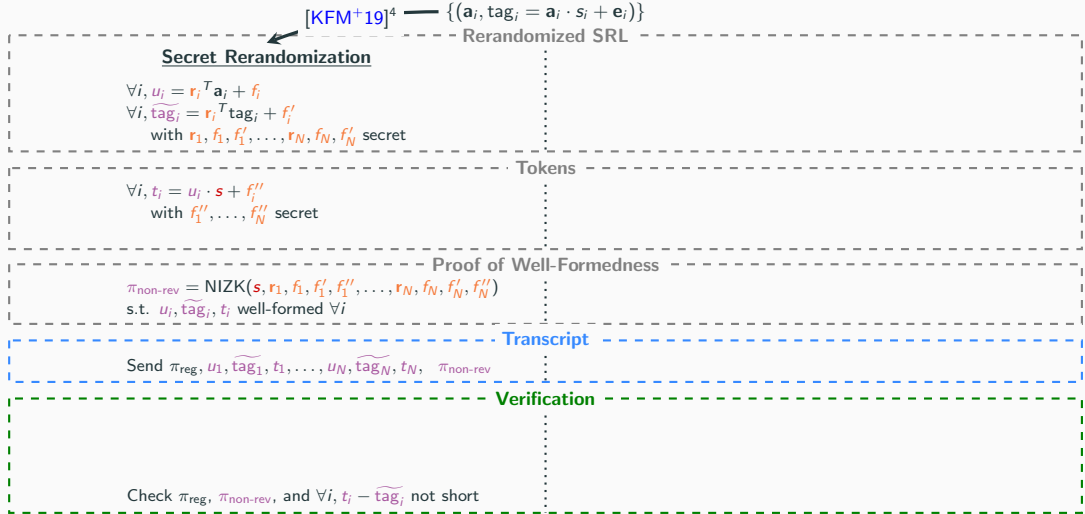
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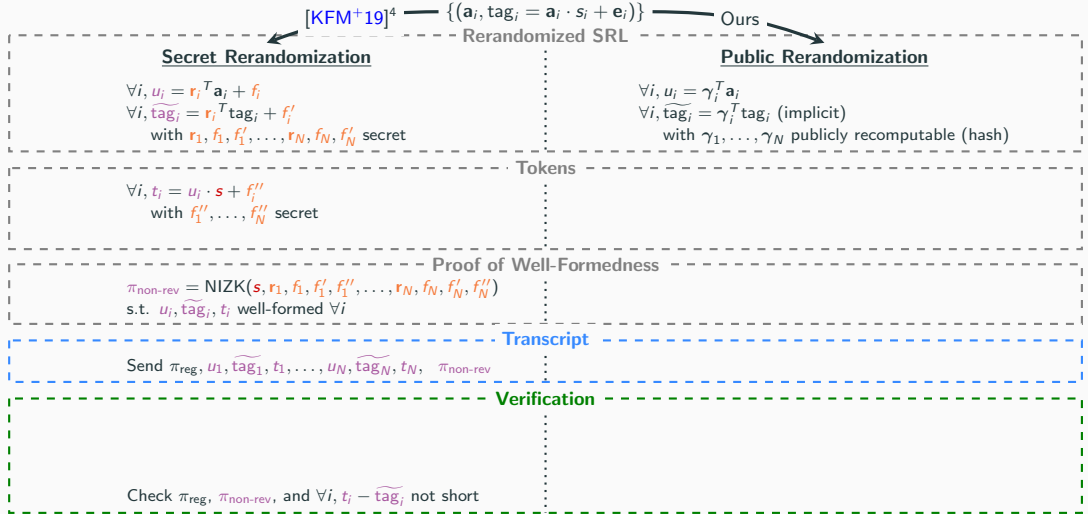
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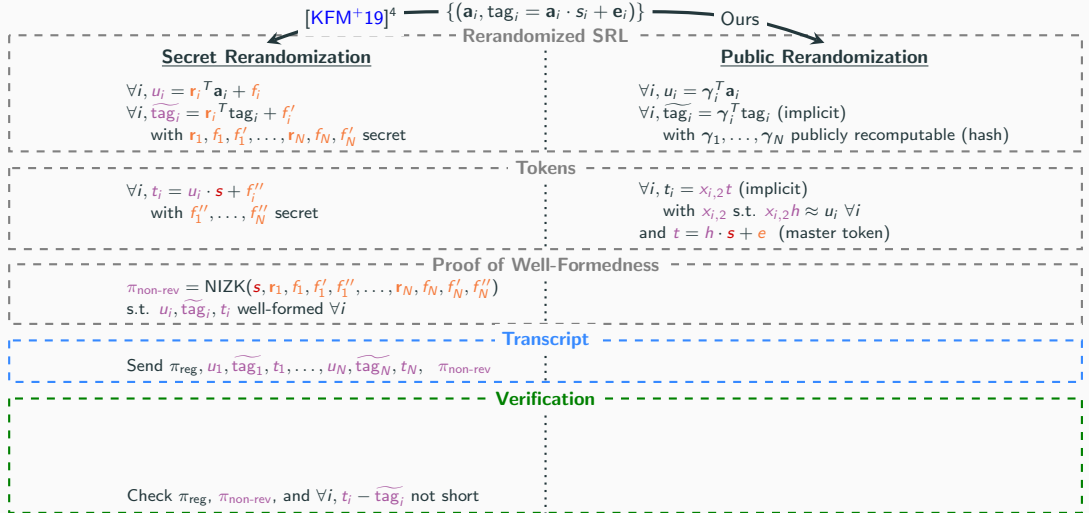
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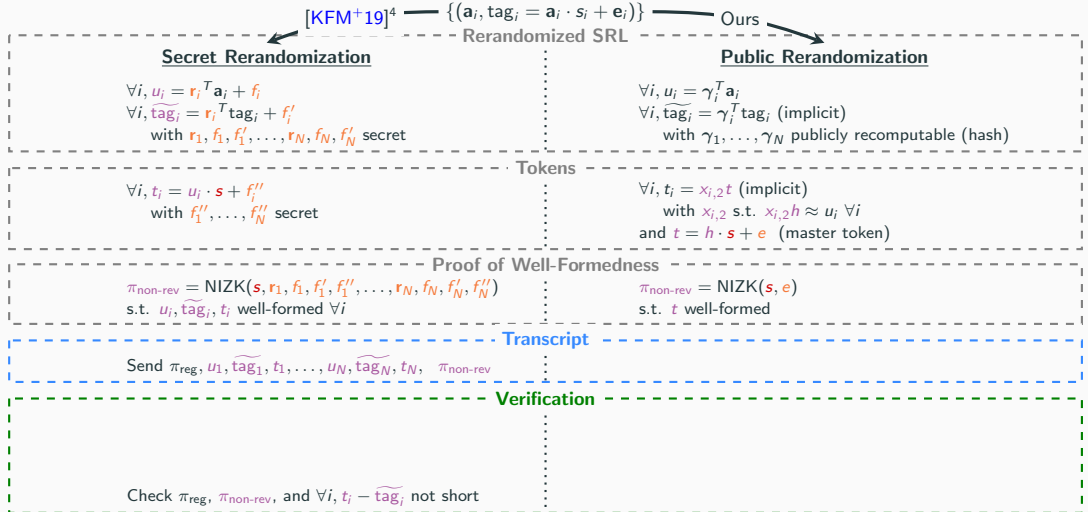
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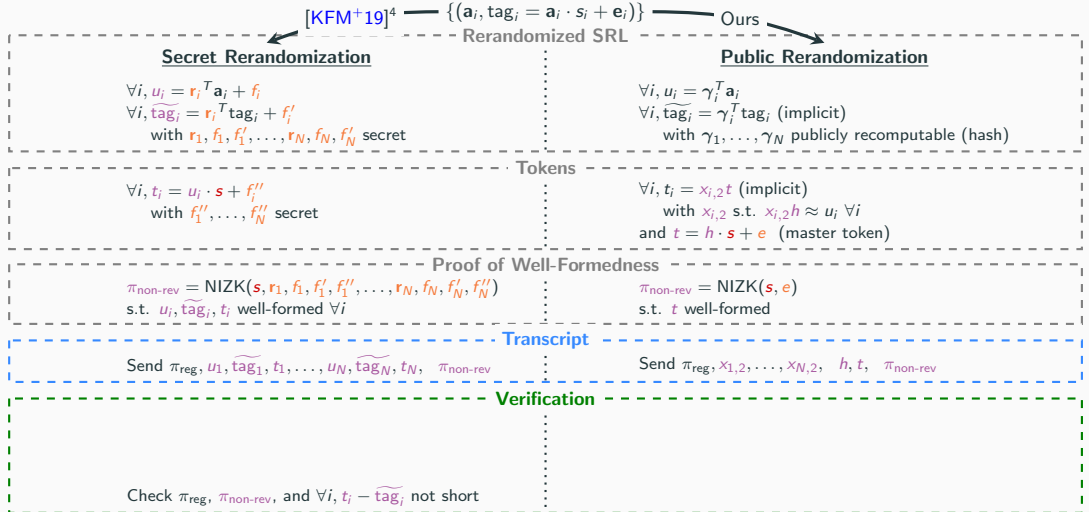
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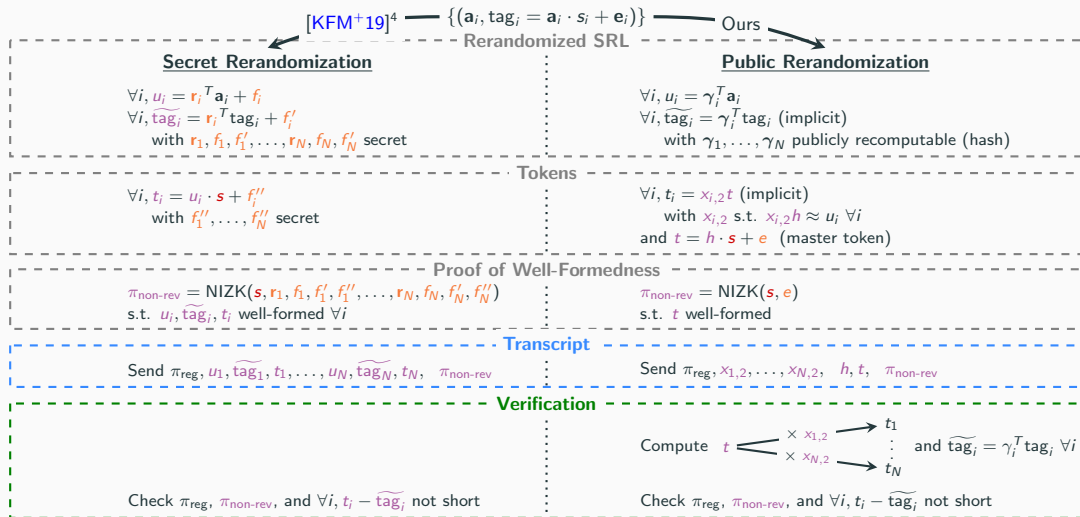
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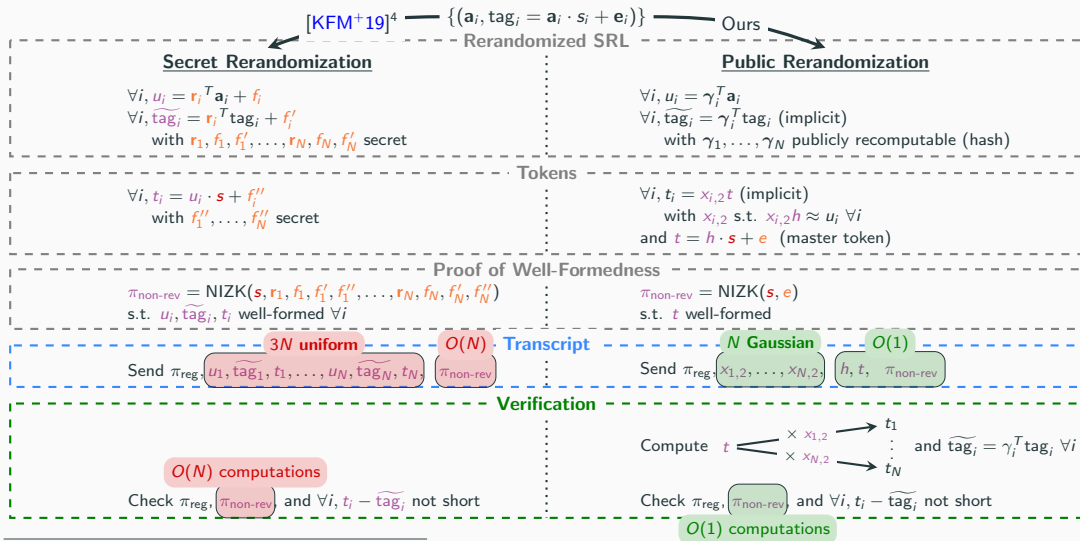
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Proving Anonymity. Need to simulate t without secret s .



Use Ring-LWE to change $hs + e$ by a uniform t .



Requires taking h perfectly uniform.



Removes the Falcon trapdoor from h , preventing generation of the $x_{i,2}$.



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Our Solution:

- (1) Program trapdoors in the $\mathbf{a}_i^T$ (ROM);
 - (2) Program preimages in the γ_i (ROM), i.e., $\mathbf{a}_i^T \gamma_i = u_i$ for u_i uniform;
 - (3) Simulate Falcon signatures and reprogram u_i , i.e., sample $x_{i,1}$, $x_{i,2}$ and define $u_i = x_{i,1} + hx_{i,2}$.
- Now h can be replaced by the uniform under NTRU (simulation of Falcon keys)



Several other security challenges for non-frameability and dealing with malicious SRL.
See full version for details (eprint 2025/1225)

Key takeaway: Security on well-known lattice assumptions (R/M-LWE, R/M-SIS, NTRU)

For a group size of 2^{32} , and $N = 1000$

	Model [†]	Assumptions	Proof System Complexity	Size
[KFM ⁺ 19]	T-SRL	Lattice	Linear NIZK	854 MB
Ours	M-SRL	Lattice	Constant NIZK	5.31 MB

[†]: T-SRL = Trusted SRL
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[BEF19] ⁵	T-SRL	Symmetric	SNARK	216.82 MB
[CDK ⁺ 24] ⁶	T-SRL	Hash-based	SNARK	11.5 MB

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New **Post-Quantum Anonymous Revocation** method for **EPID**

- › Change of paradigm with public randomization leveraging Falcon signature
- › Making the NIZK proof constant in the SRL size
- › Stronger security tackling malicious SRL, while based on standard lattice assumptions
- › Versatile anonymous revocation also suited for other privacy primitives (e.g., anonymous credentials)



Perspectives



- Combining revocation methods (e.g., ours and accumulators) to cover most scenarios at once (loss of credentials, misbehaving users, etc.) in the most efficient way



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Thank You!



D. Boneh, S. Eskandarian, and B. Fisch.

Post-quantum EPID Signatures from Symmetric Primitives.

In CT-RSA, 2019.



L. Chen, C. Dong, N. El Kassem, C. J. P. Newton, and Y. Wang.

A New Hash-Based Enhanced Privacy ID Signature Scheme.

In PQCrypto, 2024.



C. Jeudy and O. Sanders.

Worst-Case Lattice Sampler with Truncated Gadgets and Applications.

In ASIACRYPT, 2025.



N. El Kassem, L. Fiolhais, P. Martins, L. Chen, and L. Sousa.

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In WISTP, 2019.

-  V. Lyubashevsky, N. K. Nguyen, and M. Plançon.
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-  O. Sanders and J. Traoré.
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